

Coordination Compounds

Chapter 5 · All questions with full solutions



Part 1 · Examples

Q

5.1

Secondary
valence from
AgCl
precipitation
data



On the basis of the following observations made with aqueous solutions, assign secondary valences to metals in the following compounds:

Formula	Moles of AgCl precipitated per mole of the compounds with excess AgNO ₃
(i) PdCl ₂ · 4 NH ₃	2
(ii) NiCl ₂ · 6 H ₂ O	2
(iii) PtCl ₄ · 2 HCl	0
(iv) CoCl ₃ · 4 NH ₃	1
(v) PtCl ₂ · 2 NH ₃	0

Conclusion

- (i) Secondary 4
- (ii) Secondary 6
- (iii) Secondary 6
- (iv) Secondary 6
- (v) Secondary 4

Q

5.2

Writing
formulas
from IUPAC
names of
complexes



Write the formulas for the following coordination compounds:

- (a) tetraammineaquachloridocobalt(III) chloride
- (b) potassium tetrahydroxidozincate(II)
- (c) potassium trioxalatoaluminate(III)
- (d) dichloridobis(ethane-1,2-diamine)cobalt(III)
- (e) tetracarbonylnickel(0)

Part (a)

tetraammineaquachloridocobalt(III) chloride

Conclusion



Part (b)

potassium tetrahydroxidozincate(II)

Conclusion



Part (c)

potassium trioxalatoaluminate(III)

Conclusion



Part (d)

dichloridobis(ethane-1,2-diamine)cobalt(III)

Conclusion



Part (e)

tetracarbonylnickel(0)

Conclusion

$[\text{Ni}(\text{CO})_4]$

$[\text{Ni}(\text{CO})_4]$

///

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5.3

Writing
IUPAC names
of
coordination
compounds

(())

Write the IUPAC names of the following coordination compounds:

(a) $[\text{Pt}(\text{NH}_3)_2 \text{Cl}(\text{NO}_2)]$

(b) $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$

(c) $[\text{CoCl}_2(\text{en})_2] \text{Cl}$

(d) $[\text{Co}(\text{NH}_3)_5(\text{CO}_3)] \text{Cl}$

(e) $\text{Hg}[\text{Co}(\text{SCN})_4]$

Part (a)

$[\text{Pt}(\text{NH}_3)_2 \text{Cl}(\text{NO}_2)]$

Conclusion

diamminechloridonitrito-N-platinum(II)

diamminechloridonitrito-N-platinum(II)

///

Part (b)

$\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$

Conclusion

potassium trioxalatochromate(III)

potassium trioxalatochromate(III)

///

Part (c)

$[\text{CoCl}_2(\text{en})_2] \text{Cl}$

Conclusion

dichloridobis(ethane-1,2-diamine)cobalt(III) chloride

dichloridobis(ethane-1,2-diamine)cobalt(III) chloride

///

Part (d)

$[\text{Co}(\text{NH}_3)_5(\text{CO}_3)] \text{Cl}$

Conclusion

pentaamminecarbonatocobalt(III) chloride

pentaamminecarbonatocobalt(III) chloride

///

Part (e)

$\text{Hg}[\text{Co}(\text{SCN})_4]$

Conclusion

mercury (I) tetrathiocyanato-S-cobaltate(III)

mercury (I) tetrathiocyanato-S-cobaltate(III)

///

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5.4

Absence of geometrical isomerism in tetrahedral complexes



Why is geometrical isomerism not possible in tetrahedral complexes having two different types of unidentate ligands coordinated with the central metal ion?

Conclusion

Tetrahedral complexes do not show geometrical isomerism.

This is because the unidentate ligands attached to the central metal atom always have the same relative positions with respect to each other.

Q

5.5

Geometrical isomers of a mixed-ligand octahedral ion



Draw structures of geometrical isomers of $[\text{Fe}(\text{NH}_3)_2(\text{CN})_4]^-$

Conclusion

cis

trans

Q

5.6

Identifying the chiral (optically active) isomer



Out of the following two coordination entities which is chiral (optically active)?

(a) cis- $[\text{CrCl}_2(\text{ox})_2]^{3-}$

(b) trans- $[\text{CrCl}_2(\text{ox})_2]^{3-}$

Given

The two entities are represented as

(a) cis - $[\text{CrCl}_2(\text{ox})_2]^{3-}$

(b) trans - $[\text{CrCl}_2(\text{ox})_2]^{3-}$

Conclusion

Out of the two, (a) cis - $[\text{CrCl}_2(\text{ox})_2]^{3-}$ is chiral (optically active).

Q

5.7

Predicting geometry from magnetic moment

(())

The spin only magnetic moment of $[\text{MnBr}_4]^{2-}$ is 5.9 BM. Predict the geometry of the complex ion ?

Given

The coordination number of Mn^{2+} ion in the complex ion is 4. So the complex ion will be either tetrahedral ($s p^3$ hybridisation) or square planar ($d s p^2$ hybridisation).

Substitute

The magnetic moment of the complex ion is 5.9 BM. This value matches five unpaired electrons in the d orbitals. A tetrahedral shape allows five unpaired electrons, but a square planar shape does not.

Conclusion

So the complex ion is tetrahedral in shape, not square planar.

Part 2 · Additional Questions

Q

1

Formulas
from IUPAC
names



Write the formulas for the following coordination compounds:

Conclusion

- (i) $[\text{Co}(\text{H}_2\text{O})_2(\text{NH}_3)_4] \text{Cl}_3$
- (ii) $\text{K}_2[\text{Ni}(\text{CN})_4]$
- (iii) $[\text{Cr}(\text{en})_3] \text{Cl}_3$
- (vi) $[\text{Pt}(\text{NH}_3)_3 \text{BrCl}(\text{NO}_2)]^-$
- (v) $[\text{PtCl}_2(\text{en})_2](\text{NO}_3)_2$
- (vi) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$

Q

2

IUPAC names
of
coordination
compounds



Write the IUPAC names of the following coordination compounds:

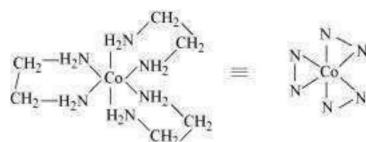
Conclusion

- (i) Hexaamminecobalt(III) chloride
- (ii) Pentaamminechloridocobalt(III) chloride
- (iii) Potassium hexacyanoferrate(III)
- (iv) Potassium trioxalatoferrate(III)
- (v) Potassium tetrachloridopalladate(II)
- (vi) Diamminechlorido(methylamine)platinum(II) chloride

Q

3

Isomerism
types and
structures



Two optical isomers of $[\text{Co}(\text{en})_3]\text{Cl}_3$

Indicate the types of isomerism exhibited by the following complexes and draw the structures for these isomers:

Given



- i. $\text{K}[\text{Cr}(\text{H}_2\text{O})_2(\text{C}_2\text{O}_4)_2]$ can exist as both geometrical (cis- and trans-) isomers. The cis-isomer can also have optical isomers.

Geometrical isomers

Trans

Cis

The trans-isomer is optically inactive, but the cis-isomer is optically active.

- (ii) Two optical isomers exist for $[\text{Co}(\text{en})_3] \text{Cl}_3$.

Two optical isomers are possible for this structure.

- (iii) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)](\text{NO}_3)_2$

A pair of optical isomers:

It can also show linkage isomerism.

Key formula

$[\text{Co}(\text{NH}_3)_5(\text{NO}_2)](\text{NO}_3)_2$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})](\text{NO}_3)_2$



Substitute

It can also show ionization isomerism.

Key formula**Conclusion**

(iv) $[\text{Pt}(\text{NH}_3)(\text{H}_2\text{O})\text{Cl}_2]$ can exist as geometrical (cis-, trans-) isomers.

Cis

Trans

Q**4**

**Evidence for
ionisation
isomerism**

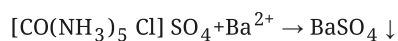
(())

Give evidence that $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Cl}$ are ionisation isomers.

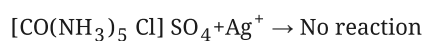
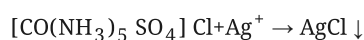
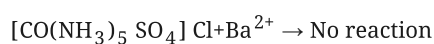
Given

When ionization isomers are dissolved in water, they ionize to give different ions.

These ions then react differently with different reagents to give different products.

**Substitute**

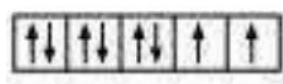
White precipitate

**Conclusion**

White precipitate

Q**5**

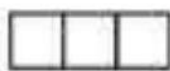
**VBT and
magnetism of
Ni complexes**



3d



4s



4p



4d



3d

4s

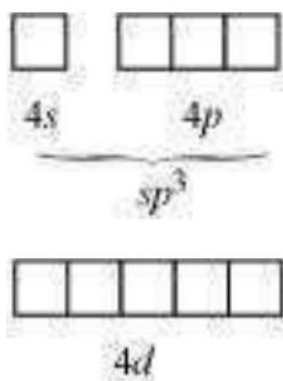
4p

4d

dsp²



3d



Explain on the basis of valence bond theory that $[\text{Ni}(\text{CN})_4]^{2-}$ ion with square planar structure is diamagnetic and the $[\text{NiCl}_4]^{2-}$ ion with tetrahedral geometry is paramagnetic.

Given

Ni is in the +2 oxidation state, i.e., in d^8 configuration.

d^8 configuration:

Substitute

There are 4 CN^- ions. So it can have either a tetrahedral or a square planar geometry. Since CN^- ion is a strong field ligand, it pairs up the unpaired 3 d electrons.

It then undergoes dsp^2 hybridization. Since all electrons are paired, it is diamagnetic. In the case of $[\text{NiCl}_4]^{2-}$, Cl^- ion, it is a weak field ligand. So it does not pair up the unpaired 3 d electrons. So it undergoes sp^3 hybridization.

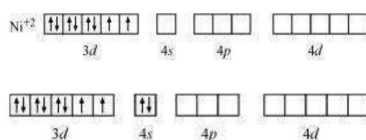
Conclusion

Since there are 2 unpaired electrons here, it is paramagnetic.

Q

6

Paramagnetism of NiCl_4 vs $\text{Ni}(\text{CO})_4$



5.6 $[\text{NiCl}_4]^{2-}$ is paramagnetic while $[\text{Ni}(\text{CO})_4]$ is diamagnetic though both are tetrahedral. Why?

Given

Though both $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CO})_4]$ are tetrahedral, their magnetic behaviour is different. This is because the ligands are different. Cl^- is a weak field ligand, so it does not pair up the unpaired 3 d electrons. So $[\text{NiCl}_4]^{2-}$ is paramagnetic.

In $\text{Ni}(\text{CO})_4$, Ni is in the zero oxidation state, i.e., it has a $3d^8 4s^2$ configuration.

Conclusion

But CO is a strong field ligand. So it pairs up the unpaired 3 d electrons. It also shifts the 4 s electrons to the 3 d orbital, giving sp^3 hybridization. Since no unpaired electrons are left, $[\text{Ni}(\text{CO})_4]$ is diamagnetic.

Q

7

Paramagnetism of aqua vs cyanido iron complexes



5.7 $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is strongly paramagnetic whereas $[\text{Fe}(\text{CN})_6]^{3-}$ is weakly paramagnetic. Explain.

Given

In both $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Fe}(\text{CN})_6]^{3-}$, Fe is in the +3 oxidation state, i.e., in d^5 configuration.



Substitute

Since CN^- is a strong field ligand, it pairs up the unpaired electrons. So only one unpaired electron is left in the d-orbital.

So,

Key formula

$$\mu = \sqrt{n(n+2)}$$



$$= \sqrt{1(1+2)}$$

$$= \sqrt{3}$$

$$= 1.732 \text{ BM}$$

Substitute

But H_2O is a weak field ligand. So it cannot pair up the electrons. This means the number of unpaired electrons is 5.

So,

Key formula

$$\mu = \sqrt{n(n+2)}$$



$$= \sqrt{5(5+2)}$$

$$= \sqrt{35}$$

$$\approx 6 \text{ BM}$$

Conclusion

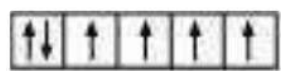
Thus, $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is strongly paramagnetic, while $[\text{Fe}(\text{CN})_6]^{3-}$ is weakly paramagnetic.

Q

8

Inner vs outer orbital complexes

(())



3d



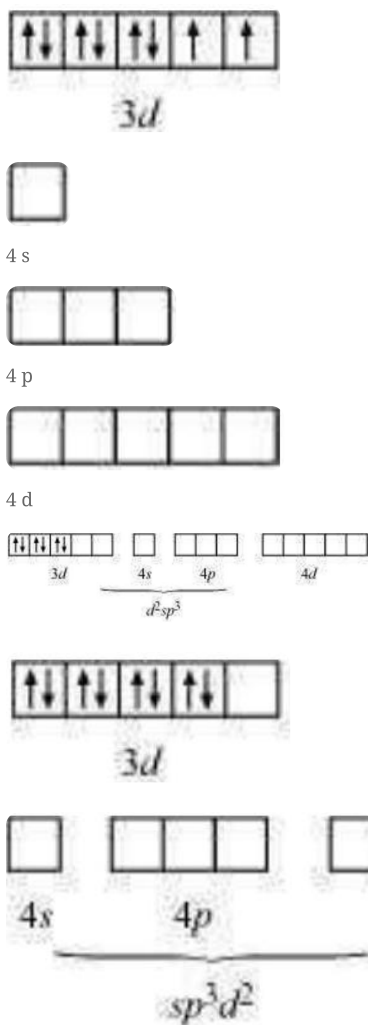
4s



4p



4d



Explain $[\text{Co}(\text{NH}_3)_6]^{3+}$ is an inner orbital complex whereas $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is an outer orbital complex.

$[\text{Co}(\text{NH}_3)_6]^{3+}$

$[\text{Ni}(\text{NH}_3)_6]^{2+}$

Oxidation state of cobalt = +3

Oxidation state of Ni = +2

Electronic configuration of cobalt = d^6

Electronic configuration of nickel = d^8

Conclusion

NH_3 being a strong field ligand causes the pairing. Therefore, Ni can undergo $d^2 s p^3$

If NH_3 causes the pairing, then only one 3 d hybridization. orbital is empty. Thus, it cannot undergo

$d^2 s p^3$ hybridization. Therefore, it undergoes $s p^3 d^2$ hybridization.

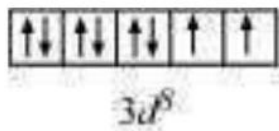
Hence, it is an inner orbital complex.

Hence, it forms an outer orbital complex.

Q

9

Unpaired
electrons in
square
planar
 $\text{Pt}(\text{CN})_4$



Predict the number of unpaired electrons in the square planar $[\text{Pt}(\text{CN})_4]^{2-}$ ion.

Given



In this complex, Pt is in the +2 state. It has a square planar structure. So it undergoes dsp^2 hybridization. The electronic configuration of $\text{Pt}(+2)$ is $5d^8$.

Conclusion

CN^- is a strong field ligand, so it pairs up the unpaired electrons. So there are no unpaired electrons in $[\text{Pt}(\text{CN})_4]^{2-}$.

Q

10

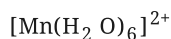
CFT and
unpaired
electrons in
Mn
complexes



The hexaquo manganese(II) ion contains five unpaired electrons, while the hexacyanoion contains only one unpaired electron.

Explain using Crystal Field Theory.

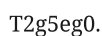
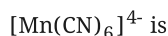
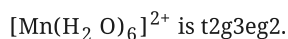
Conclusion



Mn is in the +2 oxidation state.

Mn is in the +2 oxidation state.

The electronic configuration is d^5 . The electronic configuration is d^5 . The crystal field is octahedral. Cyanide is The crystal field is octahedral. Water is a a strong field ligand. Therefore, the weak field ligand. Therefore, the arrangement of the electrons in arrangement of the electrons in



Hence, hexaquo manganese (II) ion has five unpaired electrons, while hexacyano ion has only one unpaired electron.

Q

1

Werner's postulates and bonding



Explain the bonding in coordination compounds in terms of Werner's postulates.

Given

Werner's postulates explain the bonding in coordination compounds as follows:

(i) A metal shows two types of valencies: primary and secondary.

Negative ions satisfy primary valencies.

Both negative ions and neutral molecules can satisfy secondary valencies.

Modern terminology (In modern terms, primary valency is the oxidation number of the metal ion. Secondary valency is the coordination number of the metal ion.)

Conclusion

(ii) A metal ion has a fixed number of secondary valencies around the central atom.

These valencies point in specific directions in space, giving the coordination compound its definite geometry.

(iii) Primary valencies are usually ionizable.

Secondary valencies are non-ionizable.

Q

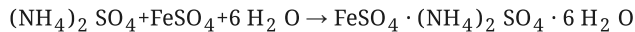
2

Complex formation vs simple ion tests

((

FeSO_4 solution mixed with $(\text{NH}_4)_2 \text{SO}_4$ solution in 1:1 molar ratio gives the test of Fe^{2+} ion but CuSO_4 solution mixed with aqueous ammonia in 1:4 molar ratio does not give the test of Cu^{2+} ion. Explain why?

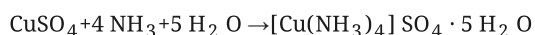
Key formula



Substitute

Mohr's salt

Key formula



Conclusion

tetraamminocopper(ii) sulphate

Both $\text{FeSO}_4 \cdot (\text{NH}_4)_2 \text{SO}_4 \cdot 6 \text{H}_2 \text{O}$ and $[\text{Cu}(\text{NH}_3)_4] \text{SO}_4 \cdot 5 \text{H}_2 \text{O}$ are addition compounds. The only major difference is that the former is a double salt, while the latter is a coordination compound.

A double salt is an addition compound. It is stable in the solid state, but breaks up into its constituent ions when dissolved. These compounds show the individual properties of their constituents. For example, $\text{FeSO}_4 \cdot (\text{NH}_4)_2 \text{SO}_4 \cdot 6 \text{H}_2 \text{O}$ breaks into Fe^{2+} , NH_4^+ and SO_4^{2-} ions. So it gives a positive test for Fe^{2+} ions. A coordination compound is an addition compound that keeps its identity both in the solid state and in solution. Here, the individual properties of the constituents are lost. This happens because $[\text{Cu}(\text{NH}_3)_4] \text{SO}_4 \cdot 5 \text{H}_2 \text{O}$ does not show the test for Cu^{2+} . The ions present in the $[\text{Cu}(\text{NH}_3)_4] \text{SO}_4 \cdot 5 \text{H}_2 \text{O}$ are $[\text{Cu}(\text{NH}_3)_4]^{2+}$ and SO_4^{2-} solution of

Q

3

Basic
coordination
compound
terminology



Explain with two examples each of the following: coordination entity, ligand, coordination number, coordination polyhedron, homoleptic and heteroleptic.

Part (i)

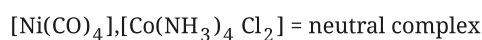
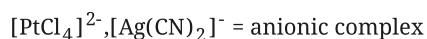
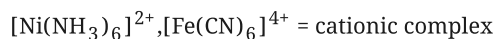
Given

Coordination entity:

A coordination entity is a charged species — it carries either a positive or a negative charge.

In a coordination entity, a suitable number of neutral molecules or negative ions (called ligands) surround the central atom or ion.

For example:



Part (ii)

Given

Ligands

Neutral molecules or negatively charged ions that surround the metal atom in a coordination entity or a coordinational complex are called ligands.

For example,

NH_3 , H_2O , Cl^- , OH^- . Ligands are usually polar and have at least one unshared pair of valence electrons.

Part (iii)

Given

Coordination number:

The coordination number of a metal atom is the total number of ligands (neutral molecules or negative ions) attached to it in the coordination sphere.

It is also called its ligancy.

For example:

(a) In the complex $\text{K}_2[\text{PtCl}_6]$, six chloride ions are attached to Pt in the coordination sphere. So the coordination number of Pt is 6.

(b) Similarly, in the complex $[\text{Ni}(\text{NH}_3)_4] \text{Cl}_2$, the coordination number of the central atom (Ni) is 4.

Part (vi)

Given

Coordination polyhedron:

The coordination polyhedron is the spatial arrangement of the ligands that are directly attached to the central metal ion in the coordination sphere.

For example:

Tetrahedral

Part (v)

Given

Homoleptic complexes:

These are complexes in which the metal ion is bound to only one kind of donor group. For example: $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{PtCl}_4]^{2-}$ etc.

Part (vi)

Given

Heteroleptic complexes:

These are complexes in which the central metal ion is bound to more than one type of donor group.

For example: $[\text{Co}(\text{NH}_3)_4 \text{Cl}_2]^+$, $[\text{Co}(\text{NH}_3)_5 \text{Cl}]^{2+}$

Q

4

Types of ligands



What is meant by unidentate, didentate and ambidentate ligands? Give two examples for each.

Part (a)

Given

Unidentate ligands: Ligands with only one donor site are called unidentate ligands.
For example, NH_3 , Cl^- etc.

Part (b)

Given

Didentate ligands: Ligands that have two donor sites are called didentate ligands.

For e.g.,

(a) Ethane-1,2-diamine

(b) Oxalate ion

Part (c)

Given

Ambidentate ligands:

Ligands that can attach themselves to the central metal atom through two different atoms are called ambidentate ligands.

For example:

(a)

(b)

Q

5

Oxidation numbers in complexes



Specify the oxidation numbers of the metals in the following coordination entities:

Part (i)



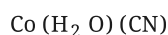
Conclusion



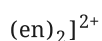
Let the oxidation number of Co be x.

The charge on the complex is +2 .

[



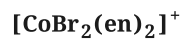
↓↓↓



$$\text{x} - 1 = +2$$

$$\text{x} = +3$$

Part (ii)



Conclusion

$$\begin{aligned} & [\text{Co}(\text{Br})_2(\text{en})_2]^{2+} \\ & \downarrow \downarrow \downarrow \\ & \cdot \\ & x + 2(-1) + 2(0) = +1 \\ & x - 2 = +1 \\ & x = +3 \end{aligned}$$

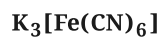
Part (iii)



Conclusion

$$\begin{aligned} & [\text{Pt}(\text{Cl})_4]^{2-} \\ & \text{Let the oxidation number of Pt be } x. \\ & \text{The charge on the complex is } -2. \\ & [\\ & \text{Pt}(\text{Cl})_4 \\ &]^{2-} \\ & \downarrow \\ & \downarrow \\ & x + 4(-1) = -2 \\ & x = +2 \end{aligned}$$

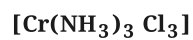
Part (iv)



Conclusion

$$\begin{aligned} & \text{K}_3[\text{Fe}(\text{CN})_6] \\ & \text{i.e., } [\\ & \text{Fe}(\text{CN})_6 \\ &]^{3-} \\ & \downarrow \downarrow \\ & x + 6(-1) = -3 \\ & x = +3 \end{aligned}$$

Part (v)



Conclusion

$$\begin{aligned} & [\\ & \text{Cr}(\text{NH}_3)_3\text{Cl}_3 \\ &] \\ & \downarrow \\ & \downarrow \end{aligned}$$

$$x+3(0)+3(-1) = 0$$

$$x-3 = 0$$

$$x = +3$$

Q

6

IUPAC
formulas
from names

(())

Using IUPAC norms write the formulas for the following:

Part (i)

Tetrahydroxidozincate(II)

Conclusion $[\text{Zn}(\text{OH})]^{2-}$

Part (ii)

Potassium tetrachloridopalladate(II)

Conclusion $\text{K}_2[\text{PdCl}_4]$

Part (iii)

Diamminedichloridoplatinum(II)

Conclusion $[\text{Pt}(\text{NH}_3)_2 \text{Cl}_2]$

Part (iv)

Potassium tetracyanonickelate(II)

Conclusion $\text{K}_2[\text{Ni}(\text{CN})_4]$

Part (v)

Pentaamminenitrito-O-cobalt(III)

Conclusion $[\text{Co}(\text{ONO})(\text{NH}_3)_5]^{2+}$

Part (vi)

Hexaamminecobalt(III) sulphate

Conclusion $[\text{Co}(\text{NH}_3)_6]_2(\text{SO}_4)_3$

Part (vii)

Potassium tri(oxalato)chromate(III)

Conclusion $\text{K}_3[\text{Cr}(\text{C}_2 \text{O}_4)_3]$

Part (viii)

Hexaammineplatinum(IV)

Conclusion $[\text{Pt}(\text{NH}_3)_6]^{4+}$

Part (ix)

Tetrabromidocuprate(II)

Conclusion $[\text{Cu}(\text{Br})_4]^{2-}$

Part (x)

Pentaamminenitrito-N-cobalt(III)

Conclusion $[\text{Co}[\text{NO}_2](\text{NH}_3)_5]^{2+}$

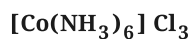
Q

7

IUPAC names
from
formulas

Using IUPAC norms write the systematic names of the following:

Part (i)

**Conclusion**

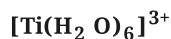
Hexaamminecobalt(III) chloride

Part (ii)

**Conclusion**

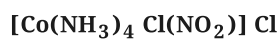
Diamminechlorido(methylamine) platinum(II) chloride

Part (iii)

**Conclusion**

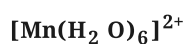
Hexaqua titanium(III) ion

Part (iv)

**Conclusion**

Tetraamminichloridonitrito-N-Cobalt(III) chloride

Part (v)

**Conclusion**

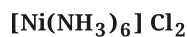
Hexaqua manganese(II) ion

Part (vi)

**Conclusion**

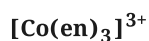
Tetrachloridonickelate(II) ion

Part (vii)

**Conclusion**

Hexaamminenickel(II) chloride

Part (viii)

**Conclusion**

Tris(ethane-1, 2-diamine) cobalt(III) ion

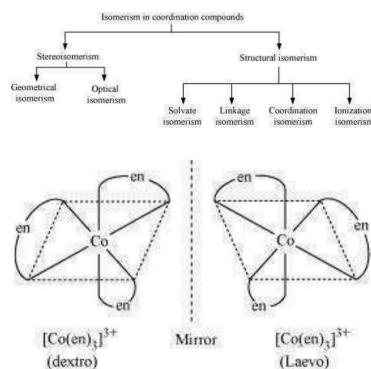
Part (ix)

**Conclusion**

Tetracarbonylnickel(0)

Q

8

Isomerism in
coordination
compounds

List various types of isomerism possible for coordination compounds, giving an example of each.

Given



(a) Geometric isomerism:

This type of isomerism is common in heteroleptic complexes.

It arises because the ligands can have different possible geometric arrangements.

For example:

Cis-isomer

Trans-isomer

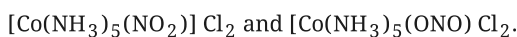
(b) Optical isomerism:

This type of isomerism arises in chiral molecules.

The isomers are mirror images of each other, and they cannot be superimposed on one another.

(c) Linkage isomerism: This type of isomerism is found in complexes that contain ambidentate ligands.

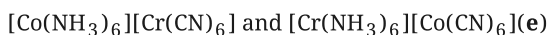
For example:



Yellow form Red form (d)

Coordination isomerism:

This type of isomerism arises when ligands are interchanged between the cationic and anionic entities of different metal ions present in the complex.

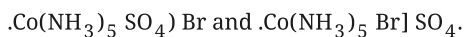


Ionization isomerism:

This type of isomerism arises when a counter ion replaces a ligand within the coordination sphere.

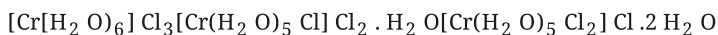
So complexes with the same composition, but that give different ions when dissolved in water, are called ionization isomers.

For example,



(f) Solvate isomerism:

Solvate isomers differ in whether the solvent molecule is directly bonded to the metal ion, or is simply present as a free solvent molecule in the crystal lattice.

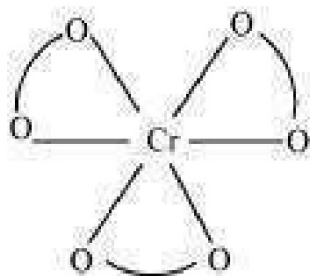


Violet Blue-green Dark green

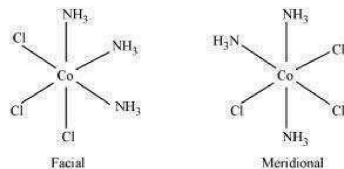
Q

9

Counting
geometrical
isomers



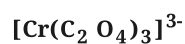
Structure of $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ - part (i), no geometrical isomer possible (bidentate ligand)



Facial and meridional geometrical isomers of $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ - part (ii)

How many geometrical isomers are possible in the following coordination entities?

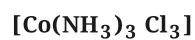
Part (i)



Conclusion

For $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$, no geometric isomer is possible as it is a bidentate ligand.

Part (ii)



Conclusion



Two geometrical isomers are possible.

Facial

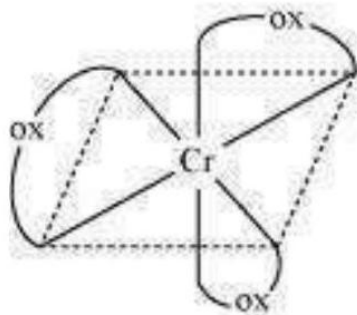
Meridional

Q

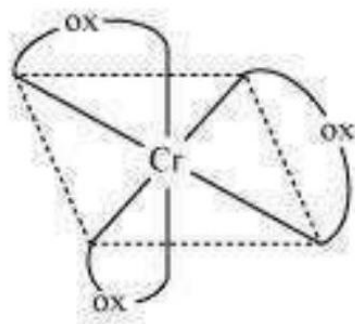
10

Optical
isomer
structures

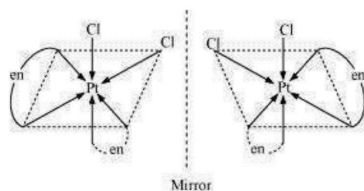
(())



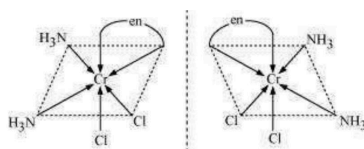
Optical isomers (dextro) of $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ - part (i)



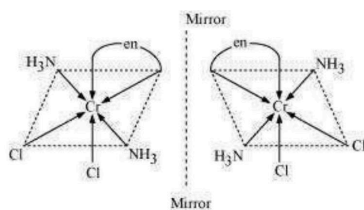
Optical isomers (laevo, mirror image) of $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ - part (i)



Optical isomers of $[\text{PtCl}_2(\text{en})_2]^{2+}$ - part (ii)



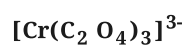
Optical isomers of $[\text{Cr}(\text{NH}_3)_2\text{Cl}_2(\text{en})]^+$ - part (iii)



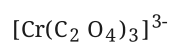
Optical isomers of $[\text{Cr}(\text{NH}_3)_2\text{Cl}_2(\text{en})]^+$ (second geometric arrangement) - part (iii)

Draw the structures of optical isomers of:

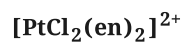
Part (i)



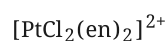
Conclusion



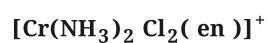
Part (ii)



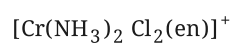
Conclusion



Part (iii)



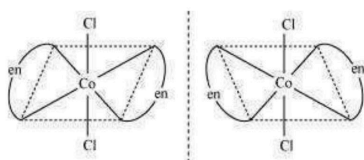
Conclusion



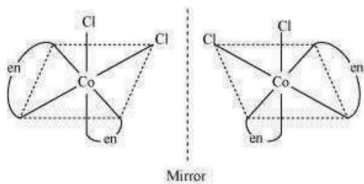
Q

11

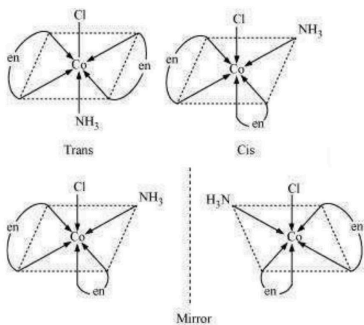
Geometrical and optical isomers



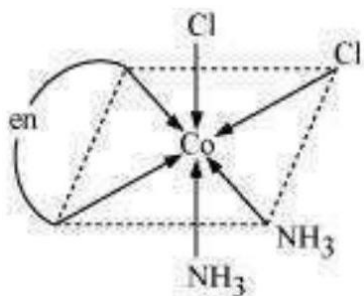
Optical isomers of $\text{trans-}[\text{CoCl}_2(\text{en})_2]^+$ - part (i)



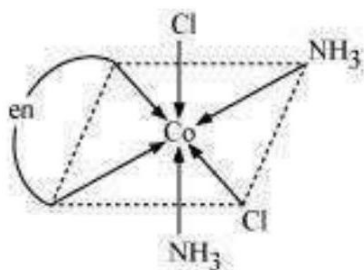
Optical isomers of $\text{cis-}[\text{CoCl}_2(\text{en})_2]^+$ - part (i)



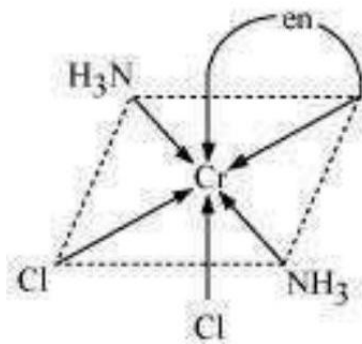
Trans and cis isomers of $[\text{Co}(\text{NH}_3)\text{Cl}(\text{en})_2]^{2+}$ - part (ii)



Optical isomer of $\text{cis-}[\text{Co}(\text{NH}_3)\text{Cl}(\text{en})_2]^{2+}$ - part (ii)



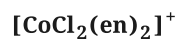
Geometrical isomer of $[\text{Co}(\text{NH}_3)_2\text{Cl}_2(\text{en})]^+$ - part (iii)



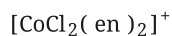
Optical isomer of $[\text{Co}(\text{NH}_3)_2\text{Cl}_2(\text{en})]^+$ -
part (iii)

Draw all the isomers (geometrical and optical) of:

Part (i)



Conclusion

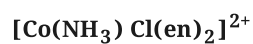


In total, three isomers are possible.

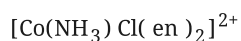
Trans $[\text{CoCl}_2(\text{en})_2]^+$ isomer - optically inactive (Superimposable mirror images).

Cis $[\text{CoCl}_2(\text{en})_2]^+$ isomer - optically active (Non-superimposable mirror images).

Part (ii)



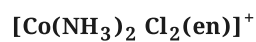
Conclusion



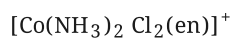
Trans-isomers are optically inactive.

Cis-isomers are optically active.

Part (iii)



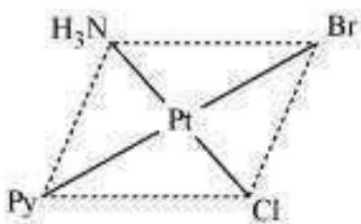
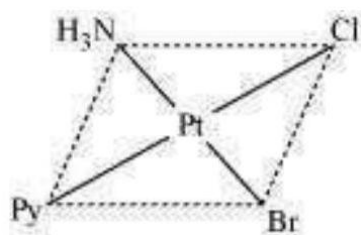
Conclusion

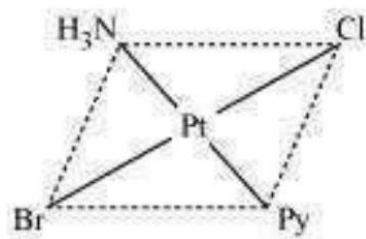


Q

12

Geometrical
and optical
isomers of a
Pt complex





Write all the geometrical isomers of $[\text{Pt}(\text{NH}_3)(\text{Br})(\text{Cl})(\text{py})]$ and how many of these will exhibit optical isomers?

Conclusion

$[\text{Pt}(\text{NH}_3)(\text{Br})(\text{Cl})(\text{py})]$.

None of the above isomers show optical isomerism.

Tetrahedral complexes rarely show optical isomerism — they do so only when unsymmetrical chelating agents are present.

Q

13

Copper sulphate reactions with KF and KCl



Aqueous copper sulphate solution (blue in colour) gives:

- (i) a green precipitate with aqueous potassium fluoride and
- (ii) a bright green solution with aqueous potassium chloride.

Explain these experimental results.

Given

Aqueous CuSO_4 exists as $[\text{Cu}(\text{H}_2\text{O})_4]\text{SO}_4$. It is blue because of the $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ ions present.

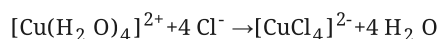


Substitute

(i) When KF is added:



(ii) When KCl is added:



(bright green)

Conclusion

In both cases, the weak field ligand water is replaced by F^- and Cl^- ions.

Q

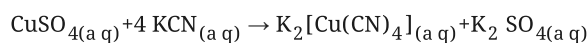
14

Copper-cyanide complex and sulphide test

(())

What is the coordination entity formed when excess of aqueous KCN is added to an aqueous solution of copper sulphate? Why is it that no precipitate of copper sulphide is obtained when $\text{H}_2\text{S}(\text{g})$ is passed through this solution?

Key formula



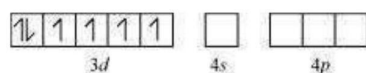
Conclusion

So the coordination entity formed is $\text{K}_2[\text{Cu}(\text{CN})_4]$. $\text{K}_2[\text{Cu}(\text{CN})_4]_{\text{is a}}$, a very stable complex. It does not ionize to give Cu^{2+} ions in water. So Cu^{2+} ions are not precipitated when $\text{H}_2\text{S}_{(\text{g})}$ is passed through the solution.

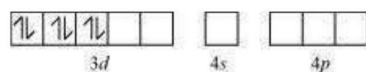
Q

15

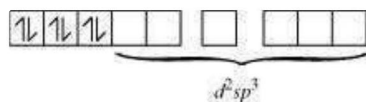
Valence bond theory of bonding



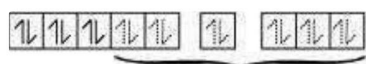
Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



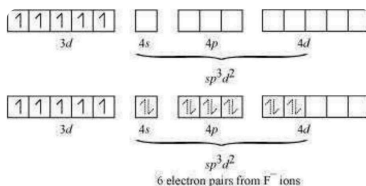
Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



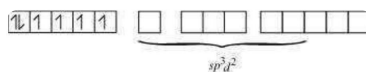
Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



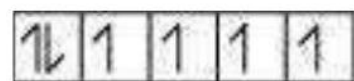
Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



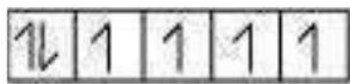
Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



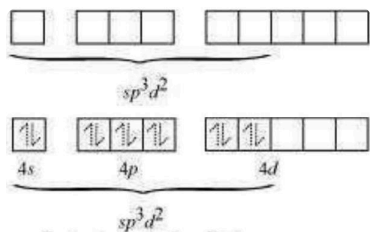
Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)



Valence bond orbital-filling diagram
(electron configuration and
hybridisation) for one of parts (i)-(iv)

Discuss the nature of bonding in the following coordination entities on the basis of valence bond theory:

Part (i)



Given



In this coordination complex, iron is in the +II oxidation state.

Fe²⁺: Electronic configuration is 3 d⁶ Orbitals

of Fe²⁺ ion:

Substitute

CN⁻ is a strong field ligand, so it causes the unpaired 3 d electrons to pair up.

There are six ligands around the central metal ion, so the most feasible hybridization is d² s p³ . d² s p³. The hybridized orbitals of Fe²⁺ are:

6 electron pairs from CN⁻ ions occupy the six hybrid d² s p³ orbitals.

Then,

Conclusion

So the geometry of the complex is octahedral, and the complex is diamagnetic (it has no unpaired electrons).

Part (ii)



Given



In this complex, the oxidation state of Fe is +3.

Orbitals of Fe⁺³ ion:

Substitute

There are 6 F⁻ ions. So it can undergo either d² s p³ or s p³ d² hybridization. F⁻ is a weak field ligand, so it does not cause pairing of the electrons in the 3 d orbital.

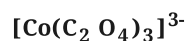
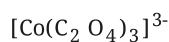
So the most feasible hybridization is s p³ d².

The s p³ d² hybridized orbitals of Fe are:

Conclusion

So the geometry of the complex is octahedral.

Part (iii)

**Given**

Cobalt is in the +3 oxidation state in this complex.

Orbitals of Co^{3+} ion:

Substitute

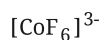
Oxalate is a weak field ligand, so it cannot cause the 3 d orbital electrons to pair up. Since there are 6 ligands, the hybridization must be either $s p^3 d^2$ or $d^2 s p^3$. $s p^3 d^2$ hybridization of Co^{3+} :

The 6 electron pairs from the 3 oxalate ions (oxalate is a bidentate ligand) occupy these $s p^3 d^2$ orbitals.

Conclusion

So the geometry of the complex is octahedral.

Part (iv)

**Given**

Cobalt is in the +3 oxidation state.

Orbitals of Co^{3+} ion:

Substitute

Fluoride ion is also a weak field ligand, so it cannot cause the 3 d electrons to pair up.

As a result, the Co^{3+} ion undergoes $s p^3 d^2$ hybridization. The $s p^3 d^2$ hybridized orbitals of Co^{3+} ion are:

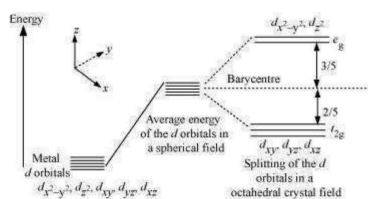
Conclusion

So the geometry of the complex is octahedral, and it is paramagnetic.

Q

16

d orbital
splitting
diagram



Draw figure to show the splitting of d orbitals in an octahedral crystal field.

Conclusion

In an octahedral field, the d orbitals split so that $d_{x^2-y^2}$ and d_{z^2} rise in energy and form the e_g level, while d_{xy} , d_{yz} and d_{zx} fall in energy and form the t_{2g} level.

Q

17

Spectrochemical series



What is spectrochemical series? Explain the difference between a weak field ligand and a strong field ligand.

Given

A spectrochemical series arranges common ligands in increasing order of their crystal-field splitting energy (CFSE) values. Ligands on the right-hand side of the series are strong field ligands. Ligands on the left-hand side are weak field ligands. Strong field ligands cause more splitting in the d orbitals than weak field ligands.

Key formula

$I < Br^- < S^{2-} < SCN^- < Cl^- < N_3^- < F^- < OH^- < C_2O_4^{2-} \approx H_2O < NCS^- \approx H^- < CN^- < NH_3$
 $< en \approx SO_3^{2-} < NO_2^- < phen < CO$

Q

18

Crystal field splitting energy

()

What is crystal field splitting energy? How does the magnitude of Δ_o decide the actual configuration of d orbitals in a coordination entity?

Given

The degenerate d-orbitals (in a spherical field environment) split into two levels, e_g and t_{2g} , when ligands are present. This splitting is called crystal-field splitting. The energy difference between the two levels (e_g and t_{2g}) is called the crystal-field splitting energy. It is denoted by Δ_o .

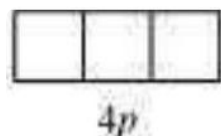
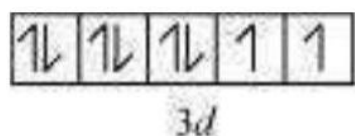
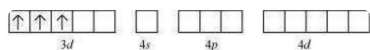
Conclusion

After the orbitals split, electrons start filling them. First, 1 electron each fills the three t_{2g} orbitals. Then the fourth electron can go in two ways: it can enter the e_g orbital, giving an $t_{2g}^3 e_g^1$ -like electronic configuration, or it can pair up in the t_{2g} orbitals, giving an $t_{2g}^4 e_g^0$ -like electronic configuration. If the Δ_o value of a ligand is less than the pairing energy (P), the electron enters the e_g orbital. If the Δ_o value of a ligand is more than the pairing energy (P), the electron enters the t_{2g} orbital.

Q

19

Paramagnetism vs diamagnetism explanation



$[Cr(NH_3)_6]^{3+}$ is paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic. Explain why?

Conclusion

Cr is in the +3 oxidation state, i.e., d^3 configuration. Also, NH_3 is a weak field ligand. It does not cause pairing of electrons in the 3 d orbital. Cr^{3+} So it undergoes $d^2 s p^3$ hybridization, and the electrons in the 3 d orbitals stay unpaired. Hence, it is paramagnetic.

In $[\text{Ni}(\text{CN})_4]^{2-}$, Ni is in the +2 oxidation state, i.e., d^8 configuration.

Ni^{2+}

CN^- is a strong field ligand. It causes pairing of the 3 d orbital electrons. Then Ni^{2+} undergoes $d s p^2$ hybridization.

As there are no unpaired electrons, it is diamagnetic.

Q

20

Colour of nickel complexes



A solution of $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is green but a solution of $[\text{Ni}(\text{CN})_4]^{2-}$ is colourless. Explain.

Conclusion

In $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$, H_2O is a weak field ligand. So there are unpaired electrons in Ni^{2+} . In this complex, the d electrons can be excited from the lower energy level to the higher energy level, i.e., d-d transition is possible. Hence, $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is coloured. In $[\text{Ni}(\text{CN})_4]^{2-}$, all the electrons are paired because CN^- is a strong field ligand. So d-d transition is not possible in $[\text{Ni}(\text{CN})_4]^{2-}$. Hence, it is colourless.

Q

21

Colour difference in iron complexes



$[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ are of different colours in dilute solutions. Why?

Conclusion

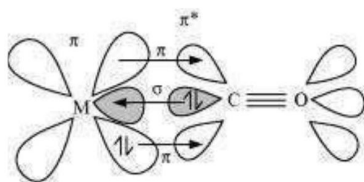
The colour of a coordination compound depends on the magnitude of the crystal-field splitting energy, Δ . This CFSE in turn depends on the nature of the ligand. In the case of $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$, the colour differs because the CFSE differs. Now, CN^- is a strong field ligand with a higher CFSE value than water. This means the energy absorbed for the intra d-d transition also differs. Hence, the transmitted colour also differs.

Q

22

Bonding in metal carbonyls

(())



Synergic bonding in metal carbonyls

Discuss the nature of bonding in metal carbonyls.

Conclusion

The metal-carbon bonds in metal carbonyls have both σ and π character. A σ bond forms when the carbonyl carbon donates a lone pair of electrons to a vacant orbital on the metal. A π bond forms when a pair of electrons from the filled metal d orbital is donated into the vacant anti-bonding π^* orbital (this is also called back bonding of the carbonyl group). The σ bond strengthens the π bond, and vice versa. This creates a synergic effect from the metal-ligand bonding, which strengthens the bond between CO and the metal.

Q

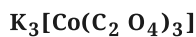
23

Oxidation
state, d-
orbital
occupation,
coordination
number

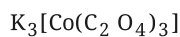


Give the oxidation state, d orbital occupation and coordination number of the central metal ion in the following complexes:

Part (i)



Given



The central metal ion is Co.

Its coordination number is 6 .

Substitute

The oxidation state can be given as:

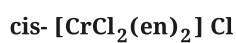
$$x - 6 = -3 \times$$

$$= +3$$

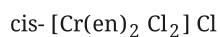
Conclusion

The d orbital occupation for Co^{3+} is $t_{2g}^6 e_g^0$.

Part (ii)



Given



The central metal ion is Cr.

The coordination number is 6 .

Substitute

The oxidation state can be given as:

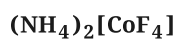
$$x + 2(0) + 2(-1) = +1$$

$$x - 2 = +1 \quad x = +3$$

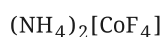
Conclusion

The d orbital occupation for Cr^{3+} is t_{2g}^3 .

Part (iii)



Given



The central metal ion is Co.

The coordination number is 4.

Substitute

The oxidation state can be given as:

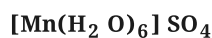
$$x - 4 = -2 \times$$

$$= +2$$

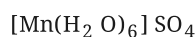
Conclusion

The d orbital occupation for Co^{2+} is $e_g^4 t_{2g}^3$.

Part (iv)



Given



The central metal ion is Mn.

The coordination number is 6 .

The

Substitute

oxidation state can be given as: $x +$

$$0 = +2x = +2$$

Conclusion

The d orbital occupation for Mn is $t_{2g}^3 e_g^2$.

Q**24**

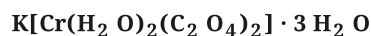
IUPAC names
and complex
properties



Write down the IUPAC name for each of the following complexes and indicate the oxidation state, electronic configuration and coordination number.

Also give stereochemistry and magnetic moment of the complex:

Part (i)

**Given**

Potassium diaquadioxalatochromate (III) trihydrate.

Oxidation state of chromium = 3

Electronic configuration: $3d^3: t_{2g}^3$

Coordination number = 6

Shape: octahedral Stereochemistry:

Trans

Cis

Trans is optically inactive

Cis is optically active

Magnetic moment, μ

Key formula

$$= \sqrt{n(n+2)}$$

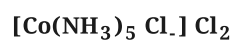
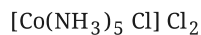
**Substitute**

$$= \sqrt{3(3+2)}$$

$$= \sqrt{15}$$

$$\approx 4 \text{ BM}$$

Part (ii)

**Conclusion**

IUPAC name: Pentaamminechloridocobalt(III) chloride

Oxidation state of Co = +3

Coordination number = 6 Shape:

octahedral.

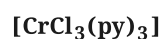
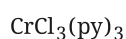
Electronic configuration $d^6: t_{2g}^6$.

Stereochemistry:

2 isomers

Magnetic Moment = 0

Part (iii)

**Given**



IUPAC name: Trichloridotripyridinechromium (III)

Oxidation state of chromium = +3

Electronic configuration for $d^3 = t_{2g}^3$

Coordination number = 6 Shape:

octahedral.

Stereochemistry:

Facial isomer

Meriodional isomer

Both isomers are optically active.

Therefore, a total of 4 isomers exist.

Key formula

Magnetic moment, $\mu = \sqrt{n(n+2)}$



Substitute

$= \sqrt{3(3+2)}$

$= \sqrt{15}$

$\approx 4 \text{ BM}$

Part (iv)

Cs[FeCl₄]

Given

Cs[FeCl₄]



IUPAC name: Caesium tetrachloroferrate (III)

Oxidation state of Fe = +3

Electronic configuration of $d^6 = e_g^2 t_{2g}^3$

Coordination number = 4

Shape: tetrahedral

Stereochemistry: optically inactive Magnetic moment:

Key formula

$= \sqrt{n(n+2)}$



$= \sqrt{5(5+2)}$

$= \sqrt{35} \approx 6 \text{ BM}$

Part (v)

K₄[Mn(CN)₆]

Given

K₄[Mn(CN)₆]



Potassium hexacyanomanganate(II)

Oxidation state of manganese = +2

Electronic configuration: $d^{5+}: t_{2g}^5$

Coordination number = 6 Shape:

octahedral.

Stereochemistry: optically inactive

Key formulaMagnetic moment, $\mu = \sqrt{n(n+2)}$ **Substitute**

$$= \sqrt{1(1+2)}$$

$$= \sqrt{3}$$

$$= 1.732 \text{ BM}$$

Q**25**

Violet colour
of titanium
complex
(CFT)

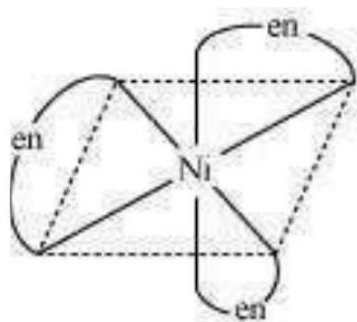


Explain the violet colour of the complex $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ on the basis of crystal field theory.

Q**26**

Chelate
effect

(())



What is meant by the chelate effect? Give an example.

Given

When a ligand attaches to a metal ion in a way that forms a ring, the metal-ligand association becomes more stable.

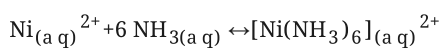


In other words, complexes with chelate rings are more stable than complexes without rings.

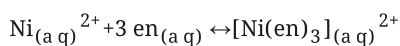
This is called the chelate effect.

Substitute

For example:

Key formula

$$\log \beta = 7.99$$



$$\log \beta = 18.1$$

(more stable)

Q

27

Applications
of
coordination
compounds



Discuss briefly giving an example in each case the role of coordination compounds in:

Part (i)

biological systems

Conclusion

Role of coordination compounds in biological systems:

Photosynthesis happens because of the chlorophyll pigment, which is a coordination compound of magnesium.

In the human body too, several coordination compounds play important roles.

For example, haemoglobin, the oxygen-carrier in blood, is a coordination compound of iron.

Part (ii)

medicinal chemistry and

Conclusion

Role of coordination compounds in medicinal chemistry:

Certain coordination compounds of platinum (for example, cis-platin) are used to inhibit the growth of tumours.

Part (iii)

analytical chemistry

Conclusion

Role of coordination compounds in analytical chemistry:

During salt analysis, many basic radicals are detected using the colour changes they show with different reagents.

These colour changes happen because the basic radicals form coordination compounds, or complexes, with different ligands.

Part (iv)

extraction / metallurgy of metals.

Conclusion

Role of coordination compounds in extraction or metallurgy of metals:

Extracting some metals from their ores involves forming complexes. For example, in aqueous solution, gold combines with cyanide ions to form $[\text{Au}(\text{CN})_2]$. Gold is later extracted from this solution by adding zinc metal.

Q

28

Ions from
cobalt
ammine
chloride in
solution



How many ions are produced from the complex $\text{Co}(\text{NH}_3)_6 \text{Cl}_2$ in solution?

(i) 6

(ii) 4

(iii) 3

(iv) 2

Conclusion

The given complex can be written as $[\text{Co}(\text{NH}_3)_6] \text{Cl}_2$.

Thus, $[\text{Co}(\text{NH}_3)_6]^+$ along with two Cl^- ions are produced.

(iii)

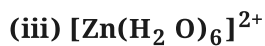
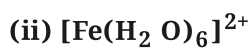
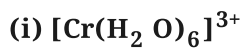
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Q

29

Highest
magnetic
moment ion

Amongst the following ions which one has the highest magnetic moment value?



Part (i)

Given

No. of unpaired electrons in $[\text{Cr}(\text{H}_2\text{O})_6]^{3+} = 3$



Key formula

Then, μ

$$= \sqrt{n(n+2)}$$



Substitute

$$= \sqrt{3(3+2)}$$

$$= \sqrt{15}$$

$$\approx 4 \text{ BM}$$

Part (ii)

Given

No. of unpaired electrons in $[\text{Fe}(\text{H}_2\text{O})_6]^{2+} = 4$



Key formula

Then, μ

$$= \sqrt{4(4+2)}$$



Substitute

$$= \sqrt{24}$$

$$\approx 5 \text{ BM}$$

Part (iii)

Conclusion

No. of unpaired electrons in $[\text{Zn}(\text{H}_2\text{O})_6]^{2+} = 0$

Answer: Hence, $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ has the highest magnetic moment value.

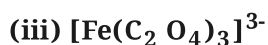
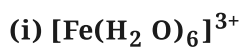
Q

30

Most stable
complex
among
options

()

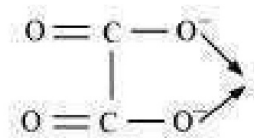
Amongst the following, the most stable complex is



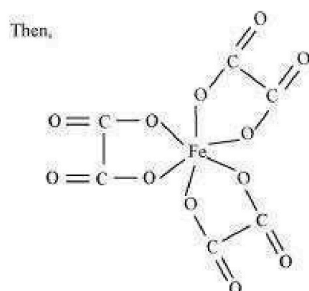
Conclusion

The stability of a complex increases with chelation. So the most stable complex is $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$.

Then,



A single oxalate (bidentate) ligand binding motif



Structure of $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$, the most stable complex, showing all three chelating oxalate ligands

Q

31

Wavelength order for nickel complexes



What will be the correct order for the wavelengths of absorption in the visible region for the following:

$[\text{Ni}(\text{NO}_2)_6]^{4-}$, $[\text{Ni}(\text{NH}_3)_6]^{2+}$, $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$?

Given

The central metal ion is the same in all three complexes.

So the absorption in the visible region depends on the ligands.

Key formula

The CFSE values of the ligands increase in the spectrochemical series in the following order:



$\text{H}_2\text{O} < \text{NH}_3 < \text{NO}_2^-$

Substitute

So the crystal-field splitting will be in the following order:

$\Delta_{(\text{H}_2\text{O})} < \Delta_{(\text{NH}_3)} < \Delta_{(\text{NO}_2^-)}$

Conclusion

Hence, the wavelengths of absorption in the visible region will be in the order:

$[\text{Ni}(\text{H}_2\text{O})_6]^{2+} > [\text{Ni}(\text{NH}_3)_6]^{2+} > [\text{Ni}(\text{NO}_2)_6]^{4-}$